INCH/POUND

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MILITARY SPECIFICATION

SPECTACLE FRAME, FRONTS AND TEMPLES, CELLULOSE ACETATE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers cellulose acetate plastic spectacle frames, fronts and temples.

1.2 Classification. Spectacle frames, fronts and temples covered by this specification shall be of the following type, class and size as specified.

Type I Frames, Male Style

Type II Fronts

Class 1 Male Style
Class 2 Female Style
Class 3 Male Style with Rocking Nose Pads

Type III Temples

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel DPSC-RST, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC/NA FSC 6540

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2. APPLICABLE DOCUMENTS.

2.1 Government Documents.

2.1.1 Specifications, Standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2)

SPECIFICATIONS

Federal

L-P-504	Plastic Sheets and Film Cellulose Acetate.
PPP-B-566	Boxes, Folding Paperboard.
PPP-B-636	Box, Shipping, Fiberboard.
PPP-B-676	Boxes, Set-up.

STANDARDS

Federal

Fed. Test Method Std. No. 151	Metals, Test Method.
Fed. Test Method Std. No. 406	Plastics, Methods of Testing.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-147	Palletized Unit Loads.
MIL-STD-794	Parts and Equipment, Procedures for Packaging of.

(Unless otherwise indicated copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS, 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

Defense Personnel Support Center (DPSC)

24442	SPECTACLE FRONT, ROCKING NOSE PADS, CELLULOSE ACETATE
24443	SPECTACLE FRONT AND TEMPLE, CELLULOSE ACETATE, MALE STYLE
24444	SPECTACLE FRONT, CELLULOSE ACETATE, FEMALE STYLE

(Application for copies should be addressed to the Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-RST, 2800 S. 20th Street, Philadelphia, PA 19101.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDiss are the issues of the documents cited in the solicitation (see 6.2).

ASTM STANDARDS

B-206	Standard Specification for Copper-Nickel-Zinc Alloy (Nickel Silver) Wire and Copper-Nickel Alloy Wire.
E-308	Standard Method for Computing the Colors of Objects by Using the CIE System.
D-569	Standard Method of Measuring the Flow Properties of Thermoplastic Molding Materials.
D-706	Standard Specification for Cellulose Acetate Molding and Extrusion Compounds.

(Applications for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Except as specified herein, material shall conform to the applicable portions of the specifications, standards and drawings specified. In addition, the metals shall conform to the chemical requirements of their applicable specifications when tested as specified in 4.4.2. When materials to be used are not specifically designated, those used shall be of the highest commercial quality, and entirely suitable for the purpose intended.

3.1.1 Plastic parts.

3.1.1.1 Cellulose acetate plastic sheets. Cellulose acetate sheets shall be extruded and shall conform to type II of L-P-504 for the thickness specified in the drawing, and as specified herein, and shall be entirely suitable for the production of opthalmic spectacle fronts and temples.

3.1.1.2 Cellulose acetate molding compound. The cellulose acetate material for injection molding opthalmic spectacle fronts and temples shall be in conformity with ASTM D 706, Class 2 or 3, Grade 1 or 2, H2 or H3 Flow. The material shall be 100 percent virgin, with no admixture of scrap, runners, or other previously molded material.

3.1.1.3 Hardness. The hardness of the plastic fronts and temples shall be 95 minimum on the Rockwell R Hardness Scale. The test shall be conducted as described in 4.4.3.

3.1.1.4 Color. The color of the sheet stock from which the fronts and temples are blanked and the color of the test coupons produced by injection molding (3.1.1.2) shall be black or brown, whenever specified. Brown color shall conform to Tennessee Eastman Co. Color Chip No. TENITE 8T131 or equal. It shall be translucent throughout.

3.1.1.4.1 Opacity. The finished plastic shall be opaque when black color is specified, at a thickness of 0.140 inch when tested as specified in 4.4.4. When brown color is specified, it shall be translucent throughout.

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3.1.1.5 Flow rate. The flow rate of the raw material (pellets) used to produce the sheet stock and for molding shall be H-2 or H-3, whichever will satisfactorily produce first quality spectacle components conforming to this specification (see 4.4.4).

3.1.1.6 Flammability. Plastic parts shall not burn faster than 24 seconds per inch when tested as specified in 4.4.5.

3.1.2 Metal parts. Metal for the hinges, rivets, nose pad arm and wire core shall be nickel silver conforming to composition 752 of ASTM B 206. Metal for the screws shall be nickel silver conforming to composition 792 of ASTM B 206. The temper shall be suitable for the purpose intended. The nose pad insert shall be pure nickel, rhodium plated. The nose pad washer shall be monel 400 material.

3.2 Construction

3.2.1 Type I - Frames. The frames shall consist of the front and temple designated for that stock number, assembled by means of the temple screw. The screw shall be peened or staked after insertion to prevent loosening. Materials shall conform to 3.1.1 and 3.1.2 of this specification, standards, and drawing specified herein.

3.2.2 Type II - Fronts. The fronts shall be constructed as specified herein and as shown on the drawing. Material for the fronts shall conform to the applicable portions of this specification, and the requirements of the drawing for style, design and dimensions of the specified size.

3.2.2.1 Style. The eyeshape shall conform to the specified drawing. Variations of eyeshape shall be within the range permitted by the drawing. Variations of the horizontal and vertical aperture dimensions (B & C) within the tolerance given shall result in corresponding variation in the radii of the lens opening to maintain the required shape. The eyeshape shall comply with the test of 4.4.8, but in no case will an irregularly shaped eye aperture be acceptable.

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3.2.2.2 Front. The fronts shall be fabricated by master cam profiling, die stamping from cellulose acetate plastic sheet as specified in 3.1.1.1, or produced by the injection molding process from cellulose acetate molding compound as specified in 3.1.1.2. The color shall be as specified in 3.1.1.4. Eyewires, bridge and endpiece shall be integral parts of the front. Dimensions of the finished front shall be as shown on the drawing. Non-toleranced dimensions shall be within manufacturer's commercial standards for quality opthalmic frame.

3.2.2.3 Rigid nose pads. If the fronts are fabricated from sheet stock, the rigid nose pads shall be made from cellulose acetate sheet as specified in 3.1.1.1 and shall be same color permanently cemented to the front. They shall be smoothly blended with the front. There shall be no chatter, file marks, or surface irregularities on the pads. The back of the nose pads (side facing the lens opening) may be made with a flat surface. Only one type of rigid nose pad shall be on each front. If the fronts are produced by injection molding, the nose pads shall be molded integrally with the front. The back of the nose pad (the side facing the lens opening) may be provided with draft, to facilitate removal from the mold. However, the nose pads shall not extend into the lens opening so as to interfere with proper glazing of the front.

3.2.2.4 Rocking nose pads. The nose pad shall be clear, colorless cellulose acetate or cellulose acetate butyrate. Metal parts shall be as specified in 3.1.2. The pad arms shall be opthalmic (not safety) weight. The nose pad assemblies shall be located on the front eyewires in such a manner the pad retaining arms shall have proper curvature and shall lie substantially in the same place as the visual axis of the finished front. The pins (rivets) shall project through the eyewire but not touching the eyewire groove. The rivets shall be spun, capped, or flathead against the pad arm base. Metal parts shall have a bright finish and be free from prominent marks.

3.2.2.5 Hinges and rivets. Each front shall be provided with two front hinge members, one right hand side and one left hand side, fabricated from nickel silver as specified in 3.1.2. The hinge members shall be three-barrel construction, and shall properly mesh with the four-barrel temple hinge member of the spectacle temples to form a complete hinge. The hinge member shall be securely held in place by the means of rivets. The rivets shall be spun, capped or flatheaded against the hinge plates. Hinges shall be polished, having a bright finish, and shall be free of prominent marks, polishing compound, sharp burrs, and dirt which detracts from their appearance.

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3.2.3 Type III - Temples. Temples shall be constructed as specified herein and as shown on the drawing. Material for the temples shall be as specified in 3.1.1. Unless otherwise specified, the temples shall conform to the requirements of the drawing for style, design, and dimensions for the specified size.

3.2.3.1 Temple assembly. The temples shall be fabricated by master cam profiling, die cutting, or die stamping from cellulose acetate plastic sheet as specified in 3.1.1.1, or produced by the injection molding process from cellulose acetate molding compound as specified in 3.1.1.2. The color shall be as specified in 3.1.1.4.

3.2.3.1.1 Temples made from sheet stock shall be made either by the lamination of two sheets of equal thickness of sheet stock (3.1.1.1) with nickel silver wire core as specified in 3.1.2, which shall be cemented or driven into the approximate center of the temple as shown in the drawing, or from solid (single thickness) sheet stock (3.1.1.1) with a nickel silver wire core as specified in 3.1.2, which shall be driven into the approximate center of the temple as shown in the drawing. As an alternate construction, the flat portion of the wire core may extend to the mitered end of the temple in lieu of the construction shown in the drawing. The flat end of the core shall be flush with mitered end of the temple. Construction used shall comply with 3.2.3.2.

3.2.3.1.2 Temples made by the injection molding process from molding compound specified in 3.1.1.2 shall have a nickel silver wire core (3.1.2) which shall be driven or molded into the approximate center of the temple as shown in the drawing. As an alternate construction, the flat portion of thee wire core may extend to the mitered end of the temple in lieu of the construction shown in the drawing. The flat end of thee core shall be flush with the mitered end of the temple. Construction used shall comply with 3.2.3.2.

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3.2.3.2 Core centering requirements. Upon visual inspection of the front edge, the core shall have a definite covering of plastic or when using alternate construction of 3.2.3.1.1 or 3.2.3.1.2 shall be visible and flush with the mitered end, the flat portion shall be parallel to the sides of the temple. If non-parallelism is observed, the angle of the edge of the core flat with the sides of the temple shall be a maximum of ten (10) degrees. No fractures, crazing, bulging, or other visible defects caused by the proximity of the core to the surfaces of the plastic shall be permitted. In the side view, the edges of the core, along its length, shall fall within two lines; each parallel to and 0.0625 inches from the centerline of the temple shown on the drawing.

3.2.3.3 Hinges and screws. Each right and left temple shall be provided with a corresponding right or left temple hinge member of four-barrel construction. They shall be fabricated from nickel silver as specified in 3.1.2 and as shown in the drawing. The hinge member shall mesh properly with a three-barrel front hinge member of the spectacle front as shown in the drawing to form a complete hinge. The temple hinge member shall have the bottom barrel tapped to accept the hinge screw. The top barrel shall be counter-sunk to accept the head of the screw. The temple hinge member shall be positioned as shown in the drawing and shall be securely fastened by means of pin rivets. The rivets shall be spun, capped, or flatheaded against the hinge plate. The rivets must go through the core without breaking the edge of the flat. Hinges shall be polished, having a bright finish, and shall be free of prominent marks, polishing compound, sharp edges, burrs, and dirt which detract from their appearance.

3.2.3.4 Temples shall be furnished in pairs, consisting of one right and one left temple of the size specified. Finished temples shall be straight and shaped as shown on the drawing without waviness or distortion.

3.3 Identification marking.

3.3.1 Fronts. Each front shall be distinctly and neatly marked with the manufacturer's name or registered trademark, eye size, and bridge size. The markings shall be located at the inner top edge.

3.3.2 Temples. Each temple shall be marked neatly, distinctly and inconspicuously, with the manufacturer's name or registered trademark, and overall temple length, and the "joint-to-bend" measurement on the inside surface of the temple butt.

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3.3.3 Frames. Each frame shall have both the front and the temples marked as specified above.

3.4 Finish. The fronts and temples shall be polished to a high gloss finish uniformly mirror-like, by means of manual or mechanical polishing. Chemical or dip methods of polishing shall not be acceptable. The gloss of the components shall be equal or superior to that of standard samples available from the procuring activity. Fronts and temples representative of lots submitted and accepted as First Article or pre-award samples may be used as standards.

3.5 Eyewire sample tests. The finished spectacle front sample shall be subjected to the following tests.

3.5.1 Eyewire tensile strength test. The load to break each eyewire, regardless of designated eye size, shall be 250 lbs. average but no one test less than 200 lbs. The test shall be conducted as specified in 4.

3.5.2 Eyewire compression test. Eyewire shall neither fracture or break when tested as described in 4.4.8.2.

3.5.3 Glazing, and adjustability. Each front sample shall be heated and glazed at least three times under normal conditions existing at military ophthalmic laboratories used in volume production of prescription spectacles, without detrimental effects, as follows:

(a) The fronts shall be capable of being properly reglazed with no loss of lens retention ability.

(b) The fronts shall be capable of being adjusted and of retaining the adjustments.

(c) There shall be no cracks, breaks or deformation in the front during and after lens insertion.

Tests shall be conducted as specified in 4.4.9. Lens retention and adjustment shall be evaluated after the samples have cooled to room temperature.

3.5.4 Gloss retention. There shall be no significant loss in the gloss after heating as specified in 4.4.10. In case of dispute, the gloss of the front after heating shall equal or surpass that of the standard when tested as specified in 4.4.10.

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3.6 Temple sample tests. Completed temples for testing purposes shall include the metal parts.

3.6.1 Adjustability. The completed temple samples shall be heated and adjusted at least three separate times under normal conditions existing at military opthalmic laboratories used in volume production and examined as follows:

(a) The temples shall be capable of being adjusted and of retaining the adjustments. Adjustment retention shall be evaluated after the adjusted temple has cooled to room temperature.

(b) There shall be no deleterious effects such as cracking or deformation.

Tests shall be conducted as specified in 4.4.9.

3.6.2 Gloss retention. There shall be no significant loss in the gloss after heating as specified in 4.4.10. In case of dispute the gloss of the temples shall equal or surpass that of the standard when tested as specified in 4.4.10.

3.7 Shape retention test. The front shall not exhibit permanent deformation exceeding 1 mm when tested as described in 4.4.11.

3.8 Interchangeability. Fronts and temples furnished under this specification shall be interchangeable (see 6.1).

3.9 Workmanship. The finish on all spectacle components, shall be of first quality. The finish shall be a high gloss all over when viewed either individually or in small groups. The surfaces of the cellulose acetate shall be free from unsightly marks. There shall be no cracks, scratches, sharp edges, visible strains or weld lines, parting lines, knockout spots, sink marks, or other imperfections which detract from the appearance of the product or may impair strength or serviceability before and after glazing. The fronts and temples shall be clean and free of any polishing residue and dirt.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.3 Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.4 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Quality conformance inspection. The quality conformance inspection shall consist of the examination and all tests of this specification.

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4.2.1 Sample specimens (coupons) for determining physical properties of the plastic material. Sufficient samples (coupons) of the sheet stock material (3.1.1.1) or coupons made of the molding compound produced as specified in 3.1.1.2 shall be taken from each lot of material used in the manufacture of the items to determine compliance with all requirements contained in this specification. The unit of product for testing shall be the quantity necessary to perform all required tests one time each, taking into consideration the fact that, where possible, the same material may be used to perform two or more tests. In addition, the contractor shall keep on file certificates of quality or results or required tests performed by the contractor, covering the physical properties of each batch or lot of material utilized in the performance of the contract.

4.2.2 For chemical analysis of the metals. Four ounces of each lot of each metal used in the construction of the items shall be selected for chemical analysis as specified in 4.4.2.

4.2.3 End item sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The unit for examination shall be the unit of issue.

TABLE I. Sampling for examination.

	Inspection level	AQL (percent defective)
For visual examination		
Major A	II	1.0
Major B	S-3	1.0
Major A	II	4.0
Major B	S-3	4.0
Major C	S-3	6.5
For dimensional examination of hinges	2 from each lot of hinges	0

4.2.3.2 For test. Sampling for tests shall be as indicated for inspection level S-1 in accordance with MIL-STD-105, except that the acceptance number shall be zero for all sample sizes, and a minimum of five samples shall be taken. Unless otherwise specified, the lot size for sampling for tests 4.4.8 eyewire strength, 4.4.9 Glazing and adjustability, 4.4.10 Gloss retention test and 4.4.11 Shape retention test shall be the production lot.

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4.3 Examination. The fronts and temples shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with, but not restricted to, the following classification of defects. Additional defects shall be classified in accordance with MIL-STD-105.

TABLE II. Categories and defects

Categories	Defects
Major A	
101	Hinges not tightly fastened as determined by visual examination and finger manipulation.
102	Hinges damaged.
103	Cellulose acetate material not free of cracks.
104	Rivets not wholly located within flat of wire core.
105	Color not conforming to specification.
Major B	
150	Eyewire groove not within tolerance.
151	Dimension "A" on the fronts not within tolerance.
152	Dimension "B" on the fronts not within tolerance.
153	Dimension "C" on the fronts not within tolerance.
154	When assembled to approved temples or fronts, as applicable, the downward angle ($10^0 \pm 3^0$) of the temples is not within tolerance.
155	When assembled to approved temples or fronts, as applicable, the setback angle ($96^0 \pm 3^0$) of the temples is greater than specified.
156	$2\text{-}3/4^0$ Angle on butt end of temple greater than $4\text{-}3/4^0$.
157	$10^0 + 2^0$ Angle butt end of temple not within tolerance.
158	Eyewire grove step less than 0.004 inch on back side.
Minor A	
201	Metal parts not free of polishing compound, dirt, prominent marks, sharp edges, and burrs.
202	Rivets not set with a smooth cap free of burrs.
203	Cellulose acetate material not free of polishing residue and dirt.

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TABLE II. Categories and defects. (Con't)

Categories	Defects
204	Fronts or temples not marked, or not properly marked, or marking not legible.
205	Groove in eyewire not centered.
206	Chatter marks in groove of eyewire.
207	Groove of eyewire irregular.
208	Groove of eyewire chipped at edges.
209	Cellulose acetate material not polished all over to a high gloss.
210	Metal hinges not polished to a bright finish.
211	Cellulose acetate material not free of scratches, sharp edges, visible strains, weld lines, flash, parting lines, knockout spots, sink marks, surface irregularities, and other imperfections.
212	Rivets not firmly set or hinge plate not flush with the adjoining surface.
213	Temples not paired in unit package.
214	Screw slot distorted.
215	Wire core not located as specified.
216	Screw missing.
217	Recess in end of screw missing.
218	Temples misshapen, bent or twisted.
Minor B	
250	Hinge plate not imbedded in the cellulose acetate material within tolerances specified.
251	When assembled to approved temples or fronts, as applicable, the setback angle ($96^0 \pm 3^0$) of the temples is less than specified.
252	The $2\text{-}3/4^0$ angle on the butt end of temple is less than $3/4^0$.
253	Size (fitting) dimension is incorrect.
254	Overall straight length of temple (Dimension "A") not within tolerances.
255	Screw threads damaged.
256	Core length not as specified.
Minor C	
301	Thickness of cellulose acetate material not within tolerance.
302	Face bow ($2^0 \pm 2^0$) of front not within tolerance.
303	Mastoid bend (tip end) of paddle not within tolerance.
304	Down bend (ear bend) of temple paddle not within tolerance.
305	Width of temple at hinge not within tolerance.
306	Width of paddle tip not within tolerance.

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4.4 Test. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Test conditions. Samples and specimens for test shall not be heated, cooled, immersed in water, or subjected to any mechanical or chemical treatment prior to test except as specifically described herein. Conditioning shall be at $23^0 + 1^0C$ ($73.5^0 \pm 2^0F$) and 50 $\pm$ 5 percent Relative Humidity immediately prior to test.

4.4.2 Chemical analysis of the metals. The samples selected in accordance with 4.2.2 shall be analyzed in conformance with Methods 111 of 112 of Federal Test Method Standard No. 151. In case of dispute, Method 111 shall be used for referee analysis.

4.4.3 Hardness. Hardness of the finished fronts and temples shall be tested as specified in Federal Test Method Standard No. 406, Method 1081, except specimen dimension shall not apply. Hardness of test specimens (4.2.1) shall be acceptable. Certification by the manufacturer of the molding or extruding material certification is acceptable.

4.4.4 Opacity. If black plastic is specified, the opacity of a 0.140 inch which sheet or molded specimen shall be determined by use of a spectro-photometer or photmeter. Transmitted light shall be zero. ASTM Standard E 308 shall be used for guidance.

4.4.5 Flow rate. Flow rate of the plastic shall be determined as described in ASTM Standard D 569.

4.4.6 Flammability. The fronts and temples shall be tested for determination of the flame propagation rate. Eyewires and temple samples shall be marked with a scribed line one inch from the end. The sample shall be ignited by holding the end of the one inch portion at the tip of a luminous 3/4 inch Bunsen burner flame. The time of burning to the one inch mark shall be determined by a stopwatch and shall be not less than the requirement of 3.1.1.6.

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4.4.7 Eyeshape. The eyeshape of the front shall be checked with standard eye plug gauge as follows: Two flat pattern-type plug gauges shall be used. One shall be a "go" gauge made to the required shape and minimum dimensions of the rear aperture of the particular eye size being examined. The other eye plug designated "no-go" shall be the required shape and the maximum dimension of the required eye size. The "go" plug gauge shall insert with ease or pass through the front and represent the shape of the aperture. The "no-go" plug shall only be capable of insertion if slight pressure is used. The front and lot shall be rejected if the "go" gauge cannot be inserted or reveals shape differences. A few (two or three) minor gaps, up to 0.010 inch for spans up to 0.250 inch shall be acceptable if the "no-go" gauge is held firmly in place by the surrounding eyewire. The front and lot shall be rejected if the "no-go" plug gauge inserts with ease and passes through the eye aperture.

4.4.7.1 Tooling. When required, layout and tooling shall be made available for Government inspection.

4.4.8 Eyewire strength. The following test shall be conducted using an electric motor driven tensile testing machine.

4.4.8.1 Eyewire tensile test. Both eyewires shall be tested, but only one at a time. The eyewire shall be secured in the machine as shown in Figure 1 and subjected to a tensile load. The load application rate (speed of testing) shall be one inch per minute until the point of fracture. The load value shall comply with 3.5.1.

4.4.8.2 Eyewire compression test. The front, including both eyewires, shall be compressed so that the overall distance between the top and the bottom of the eyewires is reduced by 25 percent. The force shall be applied at a rate of one inch per minute. Eyewire shall neither fracture nor break.

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4.4.9 Glazing and adjustability. The frame or front shall be heated and adjusted three times by qualified personnel. When frame procurement is made, the frame test sample shall remain assembled while test is conducted. For acceptance testing, the procedure described in 4.4.10 shall be used for preparing (heating) the samples used for glazing and adjustability tests. The test lens power shall range from plano to a minimum of minus three diopter sphere or sphero-cylinder. Lenses shall be glass. The lenses shall be edged to conform to the lens shape shown on the drawing for the front size being tested. Test lenses shall be correctly centered and aligned. Size tolerance shall be within ± 0.5 mm of that specified or as required by standard ANSI Z80.5. The bevel shall conform to that obtained from laboratory edgers, usually 110^0 to 125^0. The front sample under test shall be without detrimental effects, after it has been cooled to room temperature, as specified in 3.5.3.

The completed temple samples shall be heated and adjusted as specified in 3.6.1.

4.4.10 Gloss retention test. A Zyl frame warmer shall be used for this test. The warmer shall be filled with glass beads or salt heated to $260^0F + 10^0F$. The front or temple shall be completely submerged horizontally to a depth of 3/4 + 1/4 inch for a period of 30 + 2 seconds. The test sample shall be removed from the bath and allowed to cool for 5 minutes. The test samples shall be examined for gloss retention.

4.4.11 Shape retention test. Each front tested shall have the lens removed. The eyewire of the spectacle shall be heated to a temperature of $150^0 + 3^0F$ and then stretched 2 mm from its normal shape. It shall be held in this stretched position for a period of 30 minutes at ambient temperature. The eyewire is then released and allowed to recover for 24 hours. Permanent deformation shall not exceed 1 mm.

4.5 Inspection of packaging. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING.

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.2).

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5.1.1 Level A.

5.1.1.1 Unit package. Each Type I frame and Type II front and one pair of Type III temples shall be packaged in an envelope of suitable material and design. Dimensions of the envelope shall not exceed 6-1/16 inches by 2-5/16 inches. Flap (closure) shall not be glued or sealed.

5.1.1.2 Intermediate package. Twelve unit packages containing frames, fronts or pair of temples of one type, class and size only shall be packaged in a PPP-B-676. Closure shall be as specified in the applicable box specification.

5.1.2 Level C.

5.1.2.1 Unit package. Preservation for frames, fronts and temples shall conform to the requirements of MIL-STD-794 for level C.

5.1.2.2 Intermediate package. Twelve unit packages containing frames, fronts or pair of temples of one type, class and size only, shall be packaged in accordance with the requirements of MIL-STD-794 for level C.

5.2 Packing.

5.2.1 Level A. Forty-eight intermediate packaged (576 units) containing Type I frames or Type II class 3 fronts or 96 intermediate packages (1152 units) containing Type II fronts or Type III temples of one type, class and size only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping, and waterproofing shall be as specified in the applicable box specification.

5.2.2 Level B. Forty-eight intermediate packages (576 units) containing Type I frames or Type II class 3 fronts or 96 intermediate packages (1152 units) containing Type II fronts or Type II temples of one type, class and size only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. Forty-eight intermediate packages (576 units) containing Type I frames or Type II class 3 fronts or 96 intermediate packages (1152 units) containing Type II fronts or Type III temples of one type, class and size only shall be packed in accordance with the requirements of MIL-STD-794 for level C.

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5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination exceed 250 pounds or 20 cubic feet. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 20 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 54 inches in height, 48 inches in width, and 40 inches in length. Pallet shall conform to Type IV or Type V.

5.2.5.1 Levels A and B. Frames, fronts and temples packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets as specified in 5.2.5.

5.2.5.2. Level C. Frames, fronts and temples packed as specified in 5.2.3 shall be unitized as specified in MIL-STD-794.

5.3 Marking.

5.3.1 Levels A, B and C. Each unit pack, intermediate package, exterior container and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The fronts and temples are components of the standard military spectacle frame. When assembled, the frame is used to hold prescription lenses for corrective vision. The assembled frames consist of a front and pair of temples.

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6.2 Acquisition requirements.

a. Title, number, and date of the specification.

b. Issue of DoDISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1 and 2.2).

c. Applicable National Stock Number.

d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 National Stock Number (NSN) Coverage. This specification covers the following National Stock Number:

National Stock No.	Item Identification
6540-01-146-7823 (Lead Item)	FRAME, SPECTACLE, CELLULOSE ACETATE, Brown, Paddle Temples, 46 Eye, 20 Bridge
6540-01-146-7808 (Lead Item)	FRONT, SPECTACLE, CELLULOSE ACETATE, Brown, Adjustable Rocking Nose Pads, 46 Eye, 20 Bridge
6540-01-146-4272 (Lead Item)	FRONT, SPECTACLE, CELLULOSE ACETATE, Brown, Rigid Nose Pads, 46 Eye, 20 Bridge
6540-01-146-7806 (Lead Item)	FRONT, SPECTACLE, FEMALE STYLE, CELLULOSE ACETATE, Brown, Paddle Temples, 46 Eye, 18 Bridge
6540-01-145-5227 (Lead Item)	TEMPLE, SPECTACLE, CELLULOSE ACETATE, Brown, Paddle Temples, 4-1/4 Inch
6540-01-145-5230 (Lead Item)	TEMPLE, SPECTACLE, CELLULOSE ACETATE, Riding Bow Cable Type, Brown, 6-1/4 inch
6540-01-273-1464 (Lead Item)	FRAME, SPECTACLE, CELLULOSE ACETATE, Black, Paddle Temples, 46 Eye, 20 Bridge

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National Stock No.	Item Identification
6540-01-273-1461 (Lead Item)	FRONT, SPECTACLE, CELLULOSE ACETATE, Black, Adjustable Rocking Nose Pads, 46 Eye, 20 Bridge
6540-01-272-0231 (Lead Item)	FRONT, SPECTACLE, CELLULOSE ACETATE, Black, Rigid Nose Pads, 46 Eye, 20 Bridge
6540-01-272-2072 (Lead Item)	FRONT, SPECTACLE, FEMALE STYLE, CELLULOSE ACETATE, 46 Eye, 18 Bridge
6540-01-273-1466 (Lead Item)	TEMPLE, SPECTACLE, CELLULOSE ACETATE, Black, Paddle Temples, 4 Inch
6540-01-271-7690 (Lead Item)	TEMPLE, SPECTACLE, CELLULOSE ACETATE, Riding Bow Cable Type, Black, 6-1/4 inch

6.4 Subject term (key word) listing.

Spectacle frame
Male style
Female style
Fronts and Temples
Temples
Glasses frame
Fronts
Frame

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

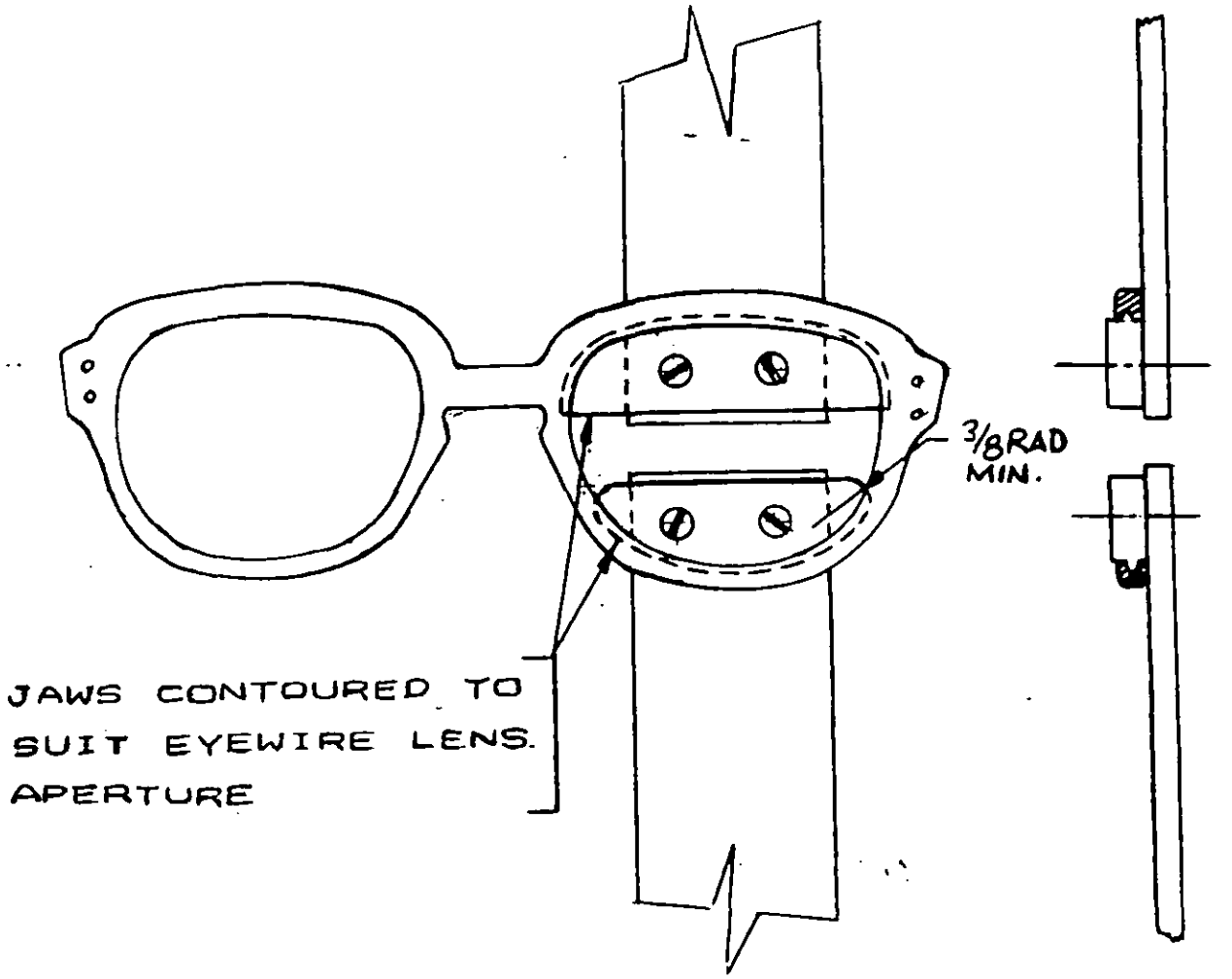
DLA-DM

CIVIL COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6540-1199

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EYEWIRE TENSILE TEST FIXTURE

FIGURE 1

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